

# A Secure and Transparent Blockchain driven E-Voting System using Smart Contract and IPFS-based Scalability

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**Abstract**—Elections are a cornerstone of democracy. Traditional voting systems such as ballots or Electronic Voting Machines face challenges such as voter fraud, reduced accessibility, security vulnerabilities, lack of transparency, and high operational costs. This paper presents a blockchain-based e-voting system that leverages InterPlanetary File System (IPFS) for secure data storage; smart contract is developed in Solidity and deployed to execute the e-voting system on the Ethereum blockchain network. Twilio-based OTP authentication is integrated to enhance security, while SHA-256 and AES-256 encryption ensure data integrity and confidentiality. The system is designed to minimize transaction costs while optimizing throughput and reducing latency. Preliminary outcomes show a significant reduction in transaction costs, increased scalability with improved system efficiency and resilience against fraudulent activities.

**Index Terms**—Blockchain, E-Voting, Smart Contracts, IPFS, Ethereum, OTP Authentication, Twilio, SHA-256, AES-256, Security, Scalability, Data Integrity, Fraud Prevention, Transparency, Digital Democracy.

## I. INTRODUCTION

The right to vote is a fundamental privilege in democracies worldwide, enabling citizens to participate in electing representatives and addressing key societal issues.

Modern voting systems remain vulnerable to hacking, tampering, and fraud, potentially compromising election integrity. Electronic voting machines (EVMs) do not fully resolve these security concerns and suffer from challenges related to universal approval [1]. Furthermore, current voting systems may lack accessibility for all voters and involve high operational costs, affecting efficiency and inclusivity.

Blockchain technology, which rose to prominence with the emergence of Bitcoin—the first cryptocurrency to achieve widespread adoption—has become a central topic in today's software industry. Blockchain stores transactions in blocks, which are linked together as new transactions occur. Each block contains transaction data, where transactions are hashed and combined into a Merkle root, stored in the block header. As of April 2025, Ethereum's market capitalization exceeds USD \$250 billion, making it the second-largest blockchain network after Bitcoin [2].

Governments worldwide are exploring blockchain to make elections tamper-proof and transparent, aiming to establish systems where data is inherently trusted and counterfeit-resistant. Blockchain offers a decentralized and secure method for recording and verifying votes, eliminating the need for intermediaries such as election officials. Estonia was the first country to establish a national Internet voting system [1].

As of April 2025, the global decentralized voting systems market is projected to grow from USD 0.35 billion in 2024 to USD 7.73 billion by 2035, with a compound annual growth rate (CAGR) of approximately 32.5% between 2025 and 2035 [3]. Blockchain's inherent properties of decentralization, transparency, and immutability provide a secure and auditable foundation for e-voting.

This paper outlines the following objectives:

- 1) **Develop a Secure Voting System:** Implement transparent smart contracts on Ethereum to automate vote management, ensuring data security, auditability, and immutability.
- 2) **Implement IPFS for Distributed Data Storage:** Utilize off-chain storage (IPFS) to minimize on-chain data, optimizing gas usage and reducing costs.
- 3) **Enhance Data Security with Cryptographic Techniques:** Integrate SHA-256 for hashing and AES-256

for encryption to protect voter identities, metadata, and vote transmissions.

- 4) **Ensure Reliable Voter Registration and Verification:** Establish a multilayered authentication mechanism using UIDAI's Aadhaar-based verification, Twilio-powered OTP authentication, and blockchain-based digital identity management to prevent impersonation and ensure that only eligible voters participate.

## II. LITERATURE SURVEY

This survey examines existing research, highlighting both its advantages and challenges.

DVTChain [1] is a blockchain-based digital voting system incorporating key entities such as the Election Commission, voters, and crypto servers within a phased voting framework. However, it lacks OTP authentication during registration and incurs high costs due to encrypted on-chain vote storage.

Pawlak et al. [4] identified persistent challenges in blockchain voting systems, including security vulnerabilities, coercion risks, cost inefficiencies, and privacy concerns. In [5], a decentralized electronic voting system was developed using the Ethereum blockchain, where smart contracts ensured secure storage of voter and candidate data on a local blockchain.

In [6], a blockchain-based e-voting system called VoteChain was designed to support large-scale, real-world polls. The system aimed to enhance election security and transparency through robust conceptualization, design, and deployment. Helios [7], a web-based, open-audit voting system, is a notable example that implements coercion-resistance mechanisms.

The integration of Aadhaar ID into e-voting was proposed in [8] to address issues such as vote duplication and tampering. This approach utilized biometric data and Virtual IDs from the Aadhaar database to enable secure voting. Electron, a centralized e-voting system developed in [9], enhanced user security by incorporating biometric sensors, effectively distinguishing between registered and unregistered users.

TrustVote [10], an advanced e-voting solution, was implemented on both public (Ethereum) and permissioned (Hyperledger) blockchains for comparative evaluation. Despite increasing interest in blockchain-based voting, organizations such as the NSA and ACM have published reports [11] cautioning against its adoption due to unresolved security and privacy concerns.

While these studies highlight the potential of blockchain-based e-voting to improve election security and transparency, challenges remain, particularly regarding scalability, privacy, and voter authentication. This research proposes a blockchain-based e-voting system designed to ensure transparency, cost-effectiveness, and accessibility, addressing key limitations identified in existing solutions.

## III. PROPOSED METHODOLOGY

The proposed system architecture consists of three primary stages: the local blockchain network, client communication, and the front-end interface. Entities (Admin, Voter, Candidate) interact with smart contracts through a Decentralized

Application (DApp), which serves as the front-end interface for decentralized devices.

Further details about each component are provided in the following sections.

### A. System Workflow

The proposed blockchain-based e-voting system employs a multi-layered architecture, as illustrated in Fig. 1. The process begins with voter registration, where users are authenticated via the Aadhaar API. Upon successful authentication, voter details are stored on IPFS; failed authentications restrict system access.

Users log in using a digital voter ID and a one-time password (OTP) for secure verification. Once authenticated, voters can generate a digital voter ID, cast their votes through smart contracts, and access election results. The administrator manages voter and candidate records, oversees election processes, and processes results.

The system ensures transparency, security, and accessibility while allowing users to view voting statistics before securely logging out.

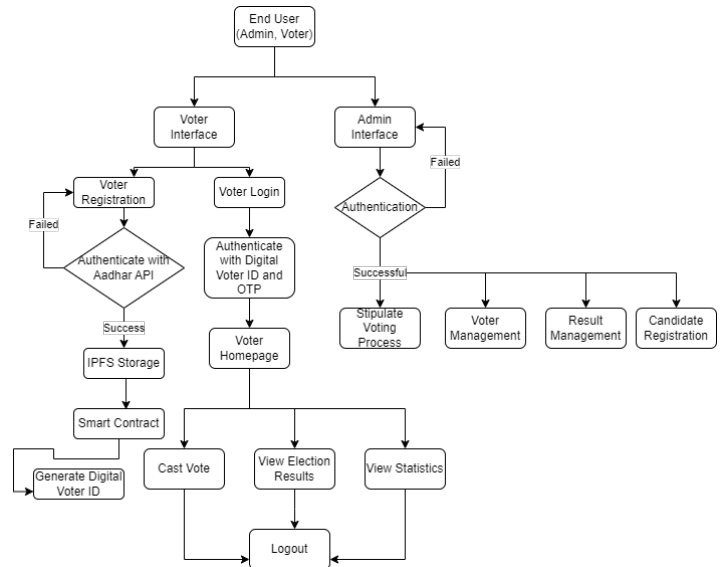


Fig. 1. System Workflow.

### B. Sequence Diagram

The implementation of the blockchain-based e-voting system follows a structured sequence to conduct the voting process, as illustrated in Fig. 2. The process is divided into the following phases:

- **Admin Registration and Candidate Enrollment:** The Election Commission registers an administrator responsible for managing the election. The administrator enlists candidates, whose details are securely stored on the blockchain.
- **Voter Registration and Digital Voter ID Generation:**
  - Voters register via a DApp frontend, providing identity details.

- Identity verification is conducted using the UIDAI Aadhaar API and OTP-based Twilio authentication.
- Upon verification, encrypted voter metadata is stored on IPFS, with SHA-256 hashing ensuring data integrity.
- The IPFS hash is stored in a smart contract, reducing gas costs while maintaining security.
- A unique *Digital Voter ID* is generated for future authentication.

#### • Voter Authentication and Login:

- Voters log in using their Digital Voter ID.
- A Twilio-based OTP authentication mechanism verifies voter identity, ensuring only legitimate users gain access.

#### • Voting Process:

- Voters access the Voting Page via the DApp frontend, ensuring the voting period has commenced before casting their vote.
- Votes are securely encrypted using AES-256 encryption and recorded on the blockchain through a smart contract.
- A confirmation message is sent to the voter via Twilio upon successful vote submission.

#### • Vote Counting and Result Declaration:

- After the voting period concludes, the system triggers the vote-counting process through smart contracts.
- Election results are securely stored on the blockchain.
- Voters and administrators can access results via the DApp interface.

#### • Election Statistics and Data Utilization:

- Data collected during voter registration and voting are utilized to generate key election statistics, including
  - \* Total voter turnout percentage,
  - \* Demographic distribution of voters,
  - \* Region-wise voting trends.
- These statistics assist in post-election analysis, comparison with previous elections, and optimization of future processes.

### C. Scalability

To scale the e-voting system for a large voter base, the following considerations are incorporated:

#### 1) Blockchain-Based Scalability:

- **Smart Contract Optimization:** Enhancing contract efficiency reduces gas costs, making transactions more affordable and scalable.
- **Sharding Implementation:** Dividing the blockchain into smaller partitions improves performance and manages high voter loads effectively.
- **Upgrading to a High-Performance Blockchain:** Networks like Ethereum 2.0 offer faster transactions and greater throughput, ensuring security and transparency.

#### 2) IPFS-Based Scalability:

- **Efficient and Scalable Storage:** Storing voter registration data on IPFS reduces blockchain storage costs and enhances system performance by preventing congestion.
- **Data Availability and Security:** Distributed storage ensures voter data remains accessible, tamper-proof, and resilient against failures or attacks.

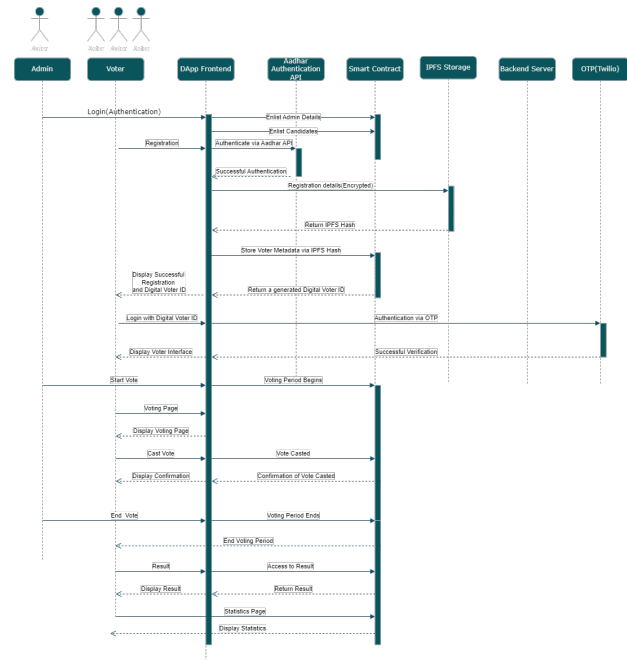


Fig. 2. Sequence Diagram of Blockchain-based E-Voting System

### D. Voter Anonymity and Privacy Preservation

To preserve voter anonymity while utilizing Aadhaar-based verification, the system employs multiple layers of security and privacy mechanisms. Aadhaar authentication is used solely during the registration phase to verify voter eligibility, and no personal information is stored on-chain. Instead, encrypted voter metadata is stored off-chain using IPFS, with only the corresponding IPFS hash stored on the blockchain.

To further enhance privacy:

- Voter metadata is encrypted using AES-256 encryption.
- The IPFS hash is hashed again using SHA-256 before being recorded on the blockchain, preventing reverse identity tracing.
- During voting, individuals interact with the system using their Digital Voter ID, acting as a pseudonym, ensuring votes remain decoupled from Aadhaar-linked identities.
- The blockchain ledger maintains only essential transaction metadata, excluding names, Aadhaar numbers, or biometric data.

This design ensures that while voter eligibility is verified, end-to-end voter anonymity is preserved throughout the election process.

## IV. IMPLEMENTATION

### A. Experimental Setup

To simulate Aadhaar authentication-based e-voting on blockchain, Ganache is used as a local Ethereum network for testing smart contracts. Solidity is the programming language for writing secure smart contracts, while Truffle serves as the development framework, offering a local blockchain, contract compiler, and testing tools. Web3.js enables interaction with the Ethereum blockchain, allowing DApp integration. Transactions use Ether (ETH) as the exchange medium, facilitated by MetaMask, a browser extension to manage Ethereum accounts and securely interact with DApp.

This setup ensures a robust and seamless development environment for blockchain-based e-voting. In our local blockchain network created in the Ganache environment, we have four dummy Ethereum wallet IDs that serve as administrators and voters, as shown in Table I. The demo accounts are credited with 100 ETHERS in each.

| Participants | Wallet Address                             |
|--------------|--------------------------------------------|
| Admin        | 0x8Dbfba37D4Da9Af7751dE0fEC3570dd860A0152  |
| Voter 1      | 0xb838807b5Fad85AE35b7EbDdd5b97030667Bd57b |
| Voter 2      | 0xD26E2277385132b0065FB06eA9bB8Ac85b0F0dba |
| Voter 3      | 0x91BaE505e39D60935B35Fc091Ef6f6dF912665A  |

TABLE I: Demo Voting Accounts with 100 ETH

### B. Design of Smart Contract

The following Solidity function allows the admin to add new candidates to the e-voting system:

Listing 1: Function for Adding New Candidates

```
1 function addCandidate(string memory _header, string
   memory _slogan)
2     public
3     // Only admin can add
4     onlyAdmin
5 {
6     Candidate memory newCandidate =
7         Candidate({
8             candidateId: candidateCount,
9             header: _header,
10            slogan: _slogan,
11            voteCount: 0
12        });
13     candidateDetails[candidateCount] = newCandidate;
14     candidateCount += 1;
15 }
```

The following Solidity function allows users to request registration as a voter:

Listing 2: Function to Register as Voter

```
1 // Function to register a voter
2 function registerAsVoter(string memory _name, string
   memory _phone) public {
3     require(!voterDetails[msg.sender].isRegistered,
4         "Voter is already registered.");
5     voterDetails[msg.sender] = Voter({
6         voterAddress: msg.sender,
7         name: _name,
8         phone: _phone,
```

```
8         isVerified: false,
9         isRegistered: true,
10        hasVoted: false
11    });
12    voterCount++;
13    voters.push(msg.sender);
14 }
```

The following Solidity function allows a verified voter to cast their vote:

Listing 3: Voting Function

```
1 // Function to cast a vote
2 function castVote(uint256 _candidateId) public {
3     require(start, "Voting has not started.");
4     require(voterDetails[msg.sender].isVerified, "
5         Voter is not verified.");
6     require(!voterDetails[msg.sender].hasVoted, "You
7         have already voted.");
8     require(!end, "Voting has ended.");
9     voterDetails[msg.sender].hasVoted = true;
10    candidateDetails[_candidateId].voteCount++;
11 }
```

The following Solidity function retrieves the total number of candidates in the system:

Listing 4: Solidity Function to Get Total Candidates

```
1 // Get candidates count
2 function getTotalCandidate() public view returns
   (uint256) {
3     // Returns total number of candidates
4     return candidateCount;
5 }
```

### C. IPFS Integration

The IPFS integration follows four key steps:

- 1) **Data Encryption:** Voter metadata is serialized into JSON, encrypted with AES-256 (CBC mode), and hashed with SHA-256 for integrity.
- 2) **IPFS Storage:** Encrypted data is uploaded to IPFS (via local node or Infura), returning a CID linked to the voter's Digital Voter ID on the Ethereum blockchain.
- 3) **Scalability with Partitioning:** IPFS's Merkle DAG structure chunks large files (default 256 KB), enabling efficient deduplication and fast retrieval. IPFS clusters with pinning ensure high availability.
- 4) **Secure Retrieval:** The DApp fetches the CID from the smart contract, retrieves and decrypts the file from IPFS, and verifies it using SHA-256 hashing.

### D. Election Statistics Generation

- **Data Retrieval:** Extract voter metadata from IPFS and voting data from smart contract event logs.
- **Preprocessing:** Parse demographics and associate voting records with Digital Voter IDs.
- **Analysis:** Calculate voter turnout, demographic trends, and candidate vote shares.
- **Visualization and Reporting:** Generate charts using Matplotlib and display summaries on the DApp frontend.

## V. RESULT AND ANALYSIS

The blockchain smart contract underwent rigorous testing, confirming its reliability and efficiency. All functions executed successfully, showcasing high system performance. The proposed e-voting system ensures confidentiality and fairness, meeting security requirements for encrypting voting data and securing transactions.

### A. Transaction and Gas Cost

The gas usage and transaction costs for different e-voting processes, including voter registration, candidate addition, election procedures, and voting process are detailed in Table II.

| Action                    | Gas (Units) | Cost (ETH) |
|---------------------------|-------------|------------|
| <b>Voter Registration</b> |             |            |
| Voter 1                   | 173800      | 0.003476   |
| Voter 2                   | 143704      | 0.002874   |
| Voter 3                   | 143668      | 0.002873   |
| <b>Add Candidates</b>     |             |            |
| Candidate 1               | 90782       | 0.001816   |
| Candidate 2               | 94838       | 0.001897   |
| Candidate 3               | 95006       | 0.001900   |
| <b>Election Process</b>   |             |            |
| Start Election            | 155212      | 0.003104   |
| Verify Voter 1            | 28697       | 0.000574   |
| Verify Voter 2            | 28697       | 0.000574   |
| Verify Voter 3            | 28697       | 0.000574   |
| End Election              | 29729       | 0.000595   |
| <b>Voting</b>             |             |            |
| Voter 1                   | 52147       | 0.001043   |
| Voter 2                   | 52159       | 0.001043   |
| Voter 3                   | 37159       | 0.000743   |

TABLE II: Gas Usage and Transaction Costs for E-Voting Process

### B. Transaction Analysis

The gas consumption across various stages of the voting system, derived from Table II, is summarized as follows:

- **Total gas cost for admin:** 551,658 units.
- **Gas costs for individual voters:**
  - Voter 1: 225,947 units
  - Voter 2: 195,863 units
  - Voter 3: 180,827 units
- **No gas consumption** for admin or voters when:
  - Viewing results, candidate lists, or admin enlistment.
- **Total gas cost for all voters:** 602,637 units.
- **Average gas expenditure per voter:** 200,879 units.
- **Overall total gas cost for the voting process:** 1,154,295 units.

### C. Simulated Latency and Throughput Analysis

To validate the scalability of the proposed blockchain-based e-voting system under a large voter base, a large-scale election simulation was conducted involving 10,000 voters and 100 candidates. The simulation was implemented on a private Ethereum blockchain using Proof of Authority (PoA) consensus, with an IPFS cluster setup to offload voter metadata

storage. Automated scripts built with Web3.js were employed to register voters, authenticate them, and cast votes at scale. Table III presents the performance metrics obtained.

| Metric                             | Value (Average)                    |
|------------------------------------|------------------------------------|
| Average Voter Registration Latency | 1.45 seconds                       |
| Average Vote Casting Latency       | 1.21 seconds                       |
| Maximum Latency Observed           | 2.98 seconds                       |
| System Throughput                  | 28.3 transactions per second (TPS) |
| Total Gas Used                     | $1.98 \times 10^9$ units           |

TABLE III: Performance Metrics of the Voting System

### D. DApp Frontend

Figure 3 showcases a screenshot of the Decentralized Application (DApp) user interface designed for the e-voting system. The DApp serves as the primary interaction layer between users and the underlying blockchain network. It provides a user-friendly, intuitive interface that allows voters to easily register, authenticate, cast votes, and view election results.

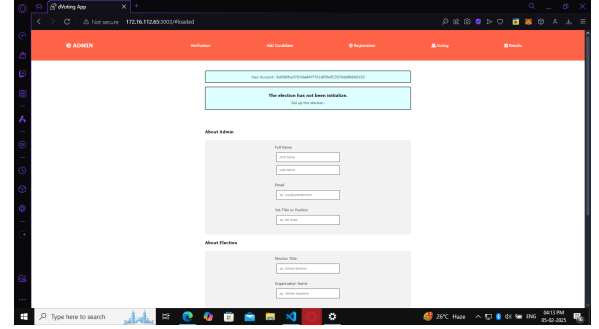


Fig. 3. HomePage of DApp for Voter

### E. Security Evaluation

To ensure the robustness of the proposed e-voting system, we define a threat model as shown in Table IV, identify potential vulnerabilities and validate the smart contract using formal analysis tool Oyente, an open-source Ethereum smart contract analyzer, to perform static analysis on the Solidity-based smart contracts. The result summary is displayed in Table V.

| Threat         | Description                      | Mitigation                                                 |
|----------------|----------------------------------|------------------------------------------------------------|
| Sybil Attack   | Fake voter identities            | Aadhaar + OTP verification; blockchain prevents duplicates |
| 51% Attack     | Control over majority of network | Uses PoA/testnets with trusted validators                  |
| DDoS Attack    | Network flooding via spam        | OTP checks + frontend/API rate limiting                    |
| Replay Attack  | Reuse of valid transactions      | Smart contract checks <code>hasVoted = true</code>         |
| IPFS Tampering | Alteration of stored metadata    | SHA-256 hash validation prevents mismatch                  |

TABLE IV: Threat Model and Mitigations

| Vulnerability Category      | Status                                |
|-----------------------------|---------------------------------------|
| Reentrancy Attack           | Not detected                          |
| Timestamp Dependence        | Safe (not used)                       |
| Integer Overflow/Underflow  | Not detected (Solidity $\geq 0.8.x$ ) |
| Unchecked Call Return Value | Properly handled                      |
| Assert/Require Misuse       | Logic validated                       |
| DoS via Revert Loops        | Not detected                          |

TABLE V: Oyente Results Summary

### F. Comparison with Related Works

Traditional EVMs are limited by centralized control, tampering risks, and scalability issues. In contrast, the blockchain system provides tamper-proof voting, scalability, transparency, and privacy, making it a more robust and future-proof solution despite its added complexity.

Table VI provides a comparative analysis of the proposed system against prominent blockchain-based e-voting frameworks. The comparison evaluates key factors such as platform, gas costs, implemented features, and privacy mechanisms. Security feature comparisons across these systems are presented in Table VII.

| Reference                       | Blockchain Platform     | Gas Usage     | Features Implemented                           | Privacy Mechanism                    |
|---------------------------------|-------------------------|---------------|------------------------------------------------|--------------------------------------|
| Alvi et al. [1]                 | Ethereum (PoW)          | Not reported  | Secure digital voting with decentralization    | Encrypted ledger, token-based access |
| TrustVote [10]                  | Ethereum + Hyperledger  | Not reported  | Dual-chain voting, analytics                   | Blockchain partitioning              |
| Khan et al. [12]                | Ethereum (PoW)          | 726,774 gas   | Vote casting                                   | Basic pseudonymity                   |
| Hjálmarsson et al. [13]         | Ethereum (PoW)          | 701,538 gas   | Vote casting, counting                         | Homomorphic encryption               |
| Dagher et al. (BroncoVote) [14] | Ethereum (PoW)          | 3,817,723 gas | Full workflow: reg., casting, tallying         | Mixnet + token-based ID              |
| <b>Proposed System</b>          | Ethereum (PoA/Test-net) | 1,154,295 gas | Aadhaar-OTP reg., voting, results, stats, IPFS | AES-256, SHA-256, IPFS               |

TABLE VI: Comparison with Existing Blockchain Voting Systems

| Features            | Khan [12] | Hjálmarsson [13] | Dagher [14] | Proposed |
|---------------------|-----------|------------------|-------------|----------|
| Anonymity           | ✗         | ✓                | ✓           | ✓        |
| Integrity           | ✓         | ✗                | ✓           | ✓        |
| Security            | ✓         | ✓                | ✓           | ✓        |
| Privacy             | ✗         | ✓                | ✓           | ✓        |
| Ind. Verifiability  | ✓         | ✓                | ✗           | ✓        |
| Univ. Verifiability | ✓         | ✓                | ✓           | ✓        |
| Mobility            | ✗         | ✓                | ✗           | ✓        |
| Uncoercibility      | ✗         | ✗                | ✓           | ✓        |

TABLE VII: Comparison of Security Features among Blockchain Voting Systems

## VI. CONCLUSION

This paper presents a secure and transparent e-voting system built on blockchain technology, integrating smart contracts, IPFS, and Aadhaar-based voter authentication to address the

key limitations of existing electoral processes. Compared to existing blockchain-based voting systems, the proposed framework offers a more complete election workflow while optimizing cost through off-chain storage and smart contract design. Future work will focus on enhancing voter privacy through the integration of privacy-preserving vote mixing techniques such as zero-knowledge proofs or blind signatures.

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